



Do Market Orientation and Supply Chain Relationship Matter in Building Innovative Capability in Trucking Business?

Kyatmaja LOOKMAN*

Doctoral Program in Management of
Technology, School of
Interdisciplinary Management and
Technology, Institut Teknologi
Sepuluh Nopember, Indonesia
kyatmaja@hotmail.com

Nyoman Pujawan

Doctoral Program in Management of
Technology, School of
Interdisciplinary Management and
Technology, Institut Teknologi
Sepuluh Nopember, Indonesia
pujawan@gmail.com

Reny Nadlifatin

Doctoral Program in Management of
Technology, School of
Interdisciplinary Management and
Technology, Institut Teknologi
Sepuluh Nopember, Indonesia
reny.nadlifatin@gmail.com

ABSTRACT

The innovative capability is a unique resource that enable companies to deliver a superior performance. It is also an essential resource to have for trucking companies, but not many studies have addressed this issue, especially in the service sector. This study investigates the factors influencing the trucking company's innovative capability, including market orientation and supply chain relationships through a case study. The factors and their indicators were obtained through extensive literature review. These factors were validated through the interview involving executives from five trucking companies. The study reveals that all the indicators obtained from the literature were considered valid and will therefore be used to develop survey instruments.

CCS CONCEPTS

• **Social and professional topics** → Professional topics.

KEYWORDS

Innovative capability, Market orientation, Supply chain relationship, Company performance

ACM Reference Format:

Kyatmaja LOOKMAN*, Nyoman Pujawan, and Reny Nadlifatin. 2022. Do Market Orientation and Supply Chain Relationship Matter in Building Innovative Capability in Trucking Business?. In *2022 The 3rd International Conference on Industrial Engineering and Industrial Management (IEIM 2022)*, January 12–14, 2022, Barcelona, Spain. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3524338.3524342>

1 INTRODUCTION

Cargo transportation is an essential component in logistics and supply chain activities. Cargo is moved by various modes, whether it is through air, sea, rail, land, and pipelines. Of the many modes, the trucking company plays a very important role because, in the end, all cargo must be delivered to the address on the land. In order to support this role, trucking companies must be market-oriented

to grow and develop. Furthermore, trucking companies are part of their customer's supply chain. Therefore good relationship with supply chain partners is a must.

The innovative capability plays a very important role in the company's future [46]. In line with the resource-based theory, the innovative capability is the company's ability to provide solutions to its customers [9][39]. For example, the ability to create, implement and develop new processes, products, and services. It will certainly make the company operate more effectively and efficiently than competitors [19], thus enabling the company to develop products and solutions to meet customer demands [31].

In order for the trucking company to be innovative, they need to be market oriented and also need to have a good relationship with their supply chain partner. Market orientation is the process of using market information to understand, anticipate and respond to current and future customer demand [5][53]. There are two types of market orientation: proactive and reactive [5][37][30]. The next factor is supply chain relationships. This factor is very important in determining the success or failure of the company [36] especially for logistics companies where the relationship occurs not only with customers but also service providers.

Furthermore, the assessment of logistic company performance is frequently done subjectively [41]. These factors influence the innovative capability of the company, which ultimately affects the company's performance. Company performance consists of financial [35][51][22][1] and non-financial [10][18][27] performances, which will be elaborated further in section two.

Many studies about innovative capabilities have been conducted in the manufacturing sectors [55][48], but few have been found in the service industry, especially in the trucking sector. Previous studies in the logistics sector have focused more on the marine transportation sector [54][24]. Of the many studies on innovative capability, not many of them have addressed the innovative capability of trucking companies. The purpose of this study is to identify factors that influence innovative capability in trucking companies. The factors and their indicators will be obtained through extensive literature review and then validated through case study.

2 LITERATURE REVIEW

2.1 Innovative Capability

The innovative capability in this research is based on resource-based theory. The innovative capability is defined as a company resources to perform innovative activities and develop solutions

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

IEIM 2022, January 12–14, 2022, Barcelona, Spain

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9569-4/22/01...\$15.00

<https://doi.org/10.1145/3524338.3524342>

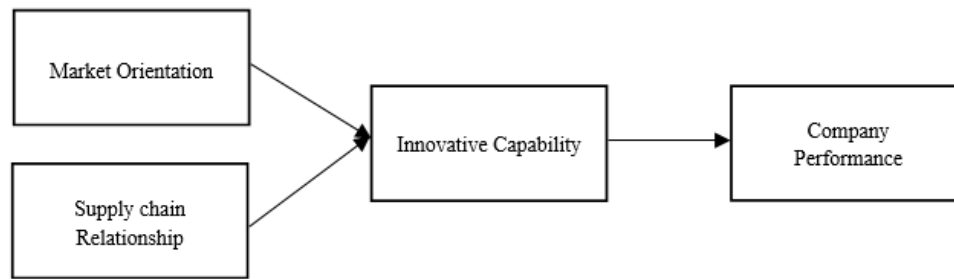


Figure 1: Conceptual model

[9][39]. The innovative capability is the company's unique resource, resulting in a superior performance according to Zhou et al. (2008) [57]. Therefore, to achieve a superior performance, companies must maximize their innovative capability to create other capabilities needed [4][49].

The innovative capability is related to the exploration and exploitation process within the company [1]. The exploration process was looking for something new with a measurable risk and aimed at finding new opportunities. On the other hand, the exploitation process maximizes the use of existing technology to find a more effective and efficient process. Both processes are important but competing for limited company resources. Successful balancing of these two processes ensures the company's competitive advantage in the future [28][34].

In this study we argue that there are two factors that influence the trucking company's innovative capability. The first is market orientation, and the second is supply chain relationships (Figure 1). Each of these factors will be further elaborated in the next subsections.

2.2 Market Orientation

The company's innovative capability is an important driver to understand better and anticipate market evolution [45]. The higher the company's willingness to make changes, the greater its market orientation. Therefore, innovative companies are also market-oriented [42], which is the first factor that affects innovative capability. Market orientation encourages companies to improve their innovative capability to remain competitive by providing better services [3] and increasing their innovation performance [2]. Market-oriented innovation capabilities will enable companies to create new offerings for their consumers [42].

There are two types of market orientation, proactive and reactive [5][37][30]. Proactive market orientation captures unknown customer demand and aims to exceed customer expectations [37]. Proactive market orientation focuses on unknown market information and seeks solutions to meet those needs. While the reactive market orientation focuses on known customer needs, companies must understand and fulfill these needs [26][37]. Customer information from responsive market orientation is easier to obtain because these needs are already known [53].

These two types of market orientation was studied in different industries, a few to none has studied market orientation in trucking company. Infact, trucking company and others have different characteristics of market orientation. Thus knowing the trucking company market orientation (proactive and reactive), will enhance the study in this field.

2.3 Supply Chain Relationship

The second factor that affects the innovative capability is supply chain relationships. The quality of these supply chain relationships constitutes social capital, which reinforces resource-based theory. The more dependent the company is on other resources, the more it is necessary to build relationships with the owners [29]. Companies are actively involved in gathering and building resources, as well as developing collective capabilities to increase shared value [20][38]. It also reduces opportunism between parties involved in the supply chain [32].

In a supply chain, good relationships have positive impacts, including building trust, relationship effectiveness [21], and reducing uncertainty, thereby reducing innovation costs [13]. The quality of the supply chain relationship increases the innovative capability [50][29]. As part of the supply chain, trucking companies are also expected to have good supply chain relationships with its supply chain partners.

2.4 Company Performance

The effect of the innovative capability on the performance of trucking companies is also investigated in this study (Figure 1). Various studies have shown a positive relationship between innovative capability and company performance [25][33], including financial performance [8][14] and ROA [44]. Previous research also shows a positive relationship with firm performance [47][40][56][6]. Yet, the innovative capability can affect performance has not been investigated in the context of trucking companies.

3 RESEARCH METHODOLOGY

This research is explorative qualitative research. This approach increases understanding of under-researched phenomena through obtaining detailed information that is difficult to capture with quantitative methods [15]. Qualitative research examines empirical evidence by interviewing people who work directly in the industry [12]. In-depth interviews were conducted with a minimum position

Table 1: Case study interview

Interview	Company	Date	Type	Duration	Job Position
I1	Company 1	26/09/2021	Online Platform	72 Min	President Director
I2	Company 2	26/09/2021	Online Platform	80 Min	President Director
I3	Company 3	26/09/2021	Online Platform	30 Min	President Director
I4	Company 4	26/09/2021	Online Platform	50 Min	Operational Director
I5	Company 5	26/09/2021	Online Olatform	40 Min	Vice Director

Table 2: Summary case study companies

Demography	Industry Served	Company Type	Truck Ownership	Yearly Income
Company 1	FMCG & Chemical (1)	Trucking Company (A)	> 100 Truck	> 25 Billion Rupiah
Company 2	Automotive (2)	Trucking Company (A)		
Company 3	Bottled Drinking Water (3)	Trucking Company (A)		
Company 4	Heavy Equipment (4)	Third-Party Logistics (B)		
Company 5	Pharmacy (5)	Distribution Company (C)		

of manager level in a trucking company. The sample was purposively chosen, because at this level, they have better understanding and authority to supply the information needed for this study. Questions and other supporting data are available and are an integral part of this study, although interviews were conducted anonymously to maintain company confidentiality. Table 1 mentions how, when and the duration the interview.

These companies were chosen based on the type of industry. In addition to being rich in the information, it also supports the research question. On top of that, it will provide an overview of the phenomena studied and increase the generalizability of this research [15]. The research design is based on the industry served by the trucking company, because the different types of industries determine the required competencies. By studying the company in the context of the industry it serves, it is hoped to generate insights and differences in that industry (table 2). The sectors served include (1) FMCG and Chemicals, (2) Automotive, (3) Bottled Drinking Water, (4) Heavy Equipment, and (5) Pharmacy. Meanwhile, the types of companies include (A) Transportation Company, (B) Third Party Logistics Company, and (C) Distribution Company.

This research method analyses the text with a textual and field approach to allocating data into a previously planned structure [16]. This method also encourages seeking clarification, especially when dominant ideas emerge during the interview. These new dominant ideas can conflict with previous findings so that participants can offer clear and complete explanations. These five case studies begin with the main question, which is then briefly presented.

Although indicators of latent variables in qualitative research are not statistically processed, qualitative research also has validity and reliability [7]. In order to achieve this, the qualitative research design must be based on a comprehensive literature review, a clear statement of the variables and indicators studied, and a strong theoretical foundation, including the innovative capability and the variables that influence and are influenced. The indicator was based on previous study to improve further validity and reliability of this research.

This study aims to determine the effect of innovative capability on company performance. The conceptual model in this study can be found in section 2. The question in this study was developed based on previous research and adapted from various sources, as can be seen in table 3. The variables of this research include market orientation (proactive and reactive), supply chain relationships, innovative capability and company performance.

4 RESULT

We found several interesting facts in the case study of five transportation companies. There are similarities and differences (table 4). The result will be discussed further in the following sub-sections.

4.1 Company 1

The company carries out a proactive market orientation by educating customers to anticipate changes in over-loading and over-dimension regulations that will take effect starting in 2023. The company offers vehicles that are larger but follow regulations. A transport management system is also offered. Currently, digitizing transportation is becoming a trend because efficiency can be increased with it. It is conveyed to the customer through a visit to the customer's office. For reactive market orientation, the company prioritizes close communication with customers. The company involves all departments to discuss current, and future customer needs. Focus on customers takes precedence over competitors through regular communication. Suggestions and input from customers are used to improve the quality of the company.

Regarding supply chain relationships, the company focuses more on customers than other partners. The company also thinks that it knows more about customer needs than other supply chain partners in terms of transportation. Meetings with supply chain partners are infrequent unless there is a new customer need involving other partners or a problem. Commitment to supply chain partners is carried out proportionally, depending on the supply chain partner's contribution to the company.

Table 3: Variable and indicator

Variable	Indicator	References
Market Orientation	1. The company helps customers to anticipate market developments	(Wang & Liu, 2020)
Proactive Market Orientation	2. The company searches for and discovers customer needs that are not yet known	
	3. The company creates solutions for customers that are not yet known to customers in the company's services	
	4. The company relates and discusses with customers to use the company's services	
Reactive Market Orientation	1. Companies measure the level of commitment and orientation to serve customers	
	2. The company discusses customer experience in all company departments	
	3. Companies focus on customers more than competitors	
	4. The company disseminates customer data to all departments at all levels regularly	
Supply Chain Relationship	1. Companies trust and collaborate with supply chain partners in the innovation process	(Li, 2020)
	2. The company communicates effectively with supply chain partners	(Bahta et al., 2020)
Innovative Capability	3. The company is fully committed to supply chain partners	
	1. The company has a culture and management that supports and encourages innovation	
	2. Companies use knowledge from various resources for product and service development more efficiently and effectively	
	3. The company supports and encourages employees to participate, generate ideas and improve the innovation process in product and service development	(Chen et al., 2020)
	4. Companies evaluate new ideas from various sources and use them in product and service development	
Company Performance	1. The company experienced an annual sales growth rate over the past three years	
	2. The company experienced a growth rate of new products over the past three years	
	3. The company is experiencing a growth rate of innovation performance	
	4. The company is experiencing growth in product performance	

Table 4: Interview result

Indicator	Company 1	Company 2	Company 3	Company 4	Company 5
Market Orientation					
Proactive Market Orientation	Yes	Yes	Yes	Yes	Yes
Reactive Market Orientation	Yes	Yes	Yes	Yes	Yes
Supply Chain Relationship	No	Yes	Yes	Yes	Yes
Innovative Capability	Yes	Yes	Yes	Yes	Yes
Company Performance	Yes	Yes	Yes	Yes	Yes

These two factors contribute to the company's innovative capability. The company develops innovative capability by nurturing a management culture that encourages innovation. This culture is formed through the courage to take risks, consistency and strong leadership. Company management also develops skills through training and education. Participation of employees from all levels is continuously encouraged. However, the selection and implementation process is carried out by upper-middle management positions. This innovative capability has caused the company to experience increased revenue, products and services in the last three years. The performance of innovation, products and services also increased.

4.2 Company 2

Proactive market orientation is carried out by the company by recording and collecting data, and then it is used to provide input on problems that may arise. The company will then send a team to discuss and provide input to the customer. Manager accounts are allocated to help find and provide solutions to customers. For reactive market orientation, the company measures Key Performance Indicators (KPI) based on customer ratings. The company ensures that employees provide services based on these KPIs. Communication is carried out routinely through the company's bureaucracy

from top to bottom. Between 85-90% of the company's focus is on consumers and the rest on competitors.

The company trusts and involves supply chain partners from start to finish, even involving customers. It is done in order to solve the problem as a whole as a single supply chain. Commitment with supply chain partners is needed because the company implements a Just in Time System (JIT). Delivery consistency is needed because the potential fines incurred are very large if it is not appropriate.

These two factors drive the company's innovative capability. The company builds innovative capability by encouraging innovation made by employees and is even given an award for the best innovation. The company involves all company departments in doing this. Companies get important information not only from inside but also outside the company. This information is obtained from competitors and even customers. The company develops new ideas regularly to drive the company's innovation. Therefore, the company experienced 15% performance growth, even though it was affected by the pandemic and closed for three months. The company is growing by offering services to other industries to reduce the impact of COVID-19. The growth of innovation performance, products and services also increased.

4.3 Company 3

The company conducts a proactive market orientation by anticipating the surge in cargo faced by customers. The company monitors this and informs customers. The company performs virtual integration with customers because the customer is the parent company. Discussions with customers regarding services are carried out intensively. For reactive market orientation, the company always measures and discusses customer experience with all company departments. Focus on customers is prioritized compared to competitors because competitors are part of the company's team to serve customers. Customer data is shared proportionally to certain sections only. The company collaborates with supply chain partners in certain areas, for example, defensive driving and safety. Although the company only functions as an operator, it remains highly committed to other supply chain partners. It is intended to ensure the smooth delivery of goods to customers.

These two factors increase the company's innovative capability. Although, innovation is limited because the company only serves certain geographically limited areas. The company innovates to improve service to customers and ensure a high level of service. The company supports employees to innovate through training and development. Innovations are derived from discoveries in the field and used for service development and improvement. Not all employees are involved in this process, only supervisor level and above. Therefore, the company experienced a growth rate of products and services over the last three years. It is done in areas that competitors do not desire. The growth of product and service innovation is also on the rise.

4.4 Company 4

The company conducts a proactive market orientation by creating integrated logistics solutions. The company provides solutions to

solve customer problems and planning. Through long-term contracts, companies can provide input and understanding of customers' business models in the future. Companies provide key performance indicators (KPIs) to customers, which are discussed at all company levels to seek new ideas. For reactive market orientation, the company measures this through the customer satisfaction index, and this is discussed and derived in the form of KPIs for each employee and incorporated into performance management development (PMD) to be measured. Although customer orientation is prioritized, competitors are also invited to collaborate. Customer data can be accessed on-demand, but not all employees have access to this data.

Partnering with supply chain partners is a must, this is because the company has few assets. Fewer assets require companies to operate more agile. Therefore, companies must collaborate with supply chain partners through long-term partnerships to become customer-driven companies.

The company's innovative capability is driven by discovering something different and new, but innovation must be accountable. Arguments regarding this must also be presented transparently. Senior management always supports innovation, but employees are expected to think about the innovation project's success. Apart from internal, input is also obtained from external consultants and competitor intelligence. Therefore, the company's performance has increased sales of products and services over the last three years. The company also experienced growth in product and service innovation product performance.

4.5 Company 5

As a customer-oriented company, helping customers anticipate market developments proactively is a must for the company. By helping customers, the company also develops its business, thereby increasing revenue for the company. The company helps customers by driving digital transformation to be more responsive to market changes, expanding their business. The company has a special department to serve customers and conduct reviews regularly. For reactive market orientation, the company measures the service level. The company focuses on its customers by paying close attention to the competition in the market. It strives to be ahead of the competition. Improvements are carried out continuously by distributing data in a limited manner to the relevant sections only.

The company believes that collaboration with supply chain partners is important. Communication is carried out regularly, be it daily or through annual events. The company is committed to growing and developing together with supply chain partners. In addition, Partners are expected to be able to follow the development of the company as well, in order to create a good supply chain ecosystem.

These two factors contribute to the company's innovative capability. The innovative capability is encouraged through the development of corporate culture to support innovation. The best ideas are contested to encourage innovation, both individually and in groups. External sources are also utilized for the development of the company's services and services. Technology development is carried out internally through a special committee to assess the idea. Therefore, the company experienced high levels of sales, products

and services over the past three years. The company also experienced growth in innovation performance, product and service performance.

5 DISCUSSION

This study provides an overview of the innovative capability in trucking companies and their effect on company performance. From the results of the interview, it was found that all companies are market-oriented. Proactive market orientation has been implemented, although in a different way. It is done by assisting and providing customer input to better understand their needs in the future by creating solutions to anticipate them. All interviewed companies also implemented reactive market orientation by focusing on customer demand and with competitors in mind. The commitment to serving customers is also measured and discussed within the company.

The company has an innovative capability, and it can be seen from the culture that supports innovation. Resources from inside and outside the company are utilized to develop the company's innovation capabilities. Internally employees are encouraged to provide ideas and are even encouraged and incentivized to do so. Therefore, market orientation affects the truck company's innovative capability. It aligns with previous research conducted in different industries such as hotels [42][3].

Companies have a different view when it comes to supply chain partners, based on their priorities. Company 1 considers that it is not important to have a strong commitment with supply chain partners, while other companies consider collaboration with supply chain partners is important. Communication with supply chain partners is carried out regularly, especially at the operational level. Although the five companies have innovative capabilities, the results of this interview show that supply chain relationships are not a contributor to the innovative capability of company 1, while for other companies are considered important.

The supply chain relationships influence the innovative capability of the company, in line with previous research [50]. The relationship between companies affect supply chain relationships on innovative capability. It explains the problem with company 1 that this weakness in innovation ability is due to the weak relationships with supply chain partners [17][11][52] especially for the transportation sector, which originates from the service industry, where the products sold are services derived from intangible knowledge [23].

On the other hand, company 2 comes from the automotive sector, which implements the just in time (JIT) system. Where customers closely monitor the company's performance. Inaccurate delivery of goods (too fast or too slow) will result in disruption of production. To ensure performance, the customer enforces claims if a party's performance causes failure in the implementation of the JIT system. Implementing this system improves coordination and close relationships in the supply chain to encourage the company's innovative capability.

The five companies showed an increase in sales growth due to the growth of new products/services over the last three years. Innovation performance also increased, accompanied by growth in

product performance [3][14]. This good company performance is generated by the company's innovative capability [43][42][19].

6 CONCLUSION

This research has provided an overview of the innovative capability in trucking companies. Two factors that influence innovative capability have also been described. All companies agree that market orientation and supply chain relationships are both important for companies. This study also indicates that trucking companies need to have innovative capability to achieve better performance. The study also able to validate all the indicators to measure market orientation, supply chain relationships, and innovative capabilities that were initially obtained through extensive literature review.

This research is a preliminary step toward testing which factors influence innovative capability in trucking companies and whether or not innovative capability has significant influence on trucking companies' performance. An important follow up from this study would be to conduct a survey that involve a large sample size to test the relationships among those factors.

REFERENCES

- [1] Acosta-Prado, J.C. *et al.* 2020. Conditions of knowledge management, innovation capability and firm performance in Colombian NTBFs: A measurement scale. *VINE Journal of Information and Knowledge Management Systems*. (2020). DOI:https://doi.org/10.1108/VJIKMS-09-2019-0142.
- [2] Alhakimi, W. and Mahmoud, M. 2020. The impact of market orientation on innovativeness: evidence from Yemeni SMEs. *Asia Pacific Journal of Innovation and Entrepreneurship*. 14, 1 (2020), 47–59. DOI:https://doi.org/10.1108/apjie-08-2019-0060.
- [3] Alnawas, I. and Hemsley-Brown, J. 2019. Market orientation and hotel performance: investigating the role of high-order marketing capabilities. *International Journal of Contemporary Hospitality Management*. 31, 4 (2019), 1885–1905. DOI:https://doi.org/10.1108/IJCHM-07-2018-0564.
- [4] Amit, R. and Schoemaker, P.J.H. 1993. Strategic Assets and Organizational Rent. *Strategic Management Journal*. 14, 1 (1993), 33–46.
- [5] Atuahene-Gima, K. 2005. Resolving the Capability – Rigidity Paradox in New Product Innovation. *Journal of Marketing*. 69, October 2005 (2005), 61–83.
- [6] Bahta, D. *et al.* 2020. Corporate social responsibility, innovation capability and firm performance: evidence from SME. *Social Responsibility Journal*. 17, 6 (2020), 840–860. DOI:https://doi.org/10.1108/SRJ-12-2019-0401.
- [7] Barratt, M. *et al.* 2011. Qualitative case studies in operations management: Trends, research outcomes, and future research implications. *Journal of Operations Management*. 29, 4 (2011), 329–342. DOI:https://doi.org/10.1016/j.jom.2010.06.002.
- [8] Bigliardi, B. 2013. The effect of innovation on financial performance: A research study involving SMEs. *Innovation: Management, Policy and Practice*. 15, 2 (2013), 245–255. DOI:https://doi.org/10.5172/impp.2013.15.2.245.
- [9] Calantone, R.J. *et al.* 2002. Learning orientation, firm innovation capability, and firm performance. *Industrial Marketing Management*. 31, 6 (2002), 515–524. DOI:https://doi.org/10.1016/S0019-8501(01)00203-6.
- [10] Cegarra-Navarro, J.G. and Dewhurst, F. 2007. Linking organizational learning and customer capital through an ambidexterity context: An empirical investigation in SMEs. *International Journal of Human Resource Management*. 18, 10 (2007), 1720–1735. DOI:https://doi.org/10.1080/09585190701570882.
- [11] Cousins, P.D. *et al.* 2011. Breakthrough scanning, supplier knowledge exchange, and new product development performance. *Journal of Product Innovation Management*. 28, 6 (2011), 930–942. DOI:https://doi.org/10.1111/j.1540-5885.2011.00854.x.
- [12] Daniel, V.M. *et al.* 2017. Innovation capabilities in services: a multi-cases approach. *Academia Revista Latinoamericana de Administración*. 30, 4 (2017), 490–507. DOI:https://doi.org/10.1108/ARLA-06-2016-0161.
- [13] Delbufalo, E. 2012. Outcomes of inter-organizational trust in supply chain relationships: A systematic literature review and a meta-analysis of the empirical evidence. *Supply Chain Management*. 17, 4 (2012), 377–402. DOI:https://doi.org/10.1108/13598541211246549.
- [14] Exposito, A. and Sanchis-Llopis, J.A. 2018. Innovation and business performance for Spanish SMEs: New evidence from a multi-dimensional approach. *International Small Business Journal: Researching Entrepreneurship*. 36, 8 (2018), 911–931. DOI:https://doi.org/10.1177/0266242618782596.

- [15] Figueiredo, P.N. and Piana, J. 2018. Innovative capability building and learning linkages in knowledge-intensive service SMEs in Brazil's mining industry. *Resources Policy*. 58, October (2018), 21–33. DOI:https://doi.org/10.1016/j.resourpol.2017.10.012.
- [16] Fiske, J. and Hartley, J. 2003. *Reading television*. Routledge, Taylor & Francis Group.
- [17] Hagedoorn, J. and Frankort, H.T.W. 2008. The gloomy side of embeddedness: The effects of overembeddedness on inter-firm partnership formation. *The Electronic Library*. 503–530.
- [18] Han, M. and Celly, N. 2008. Strategic ambidexterity and performance in international new ventures. *Canadian Journal of Administrative Sciences*. 25, 4 (2008), 335–349. DOI:https://doi.org/10.1002/cjas.84.
- [19] Hanaysha, J.R. 2020. Innovation Capabilities and Authentic Leadership: Do They Really Matter to Firm Performance? *Journal of Asia-Pacific Business*. 21, 4 (2020), 271–290. DOI:https://doi.org/10.1080/10599231.2020.1824523.
- [20] Harabi, N. 1998. Innovation through vertical relations between firms, suppliers and customers: A study of German firms. *Industry and Innovation*. 5, 2 (1998), 157–179. DOI:https://doi.org/10.1080/13662719800000009.
- [21] Henke, J.W. and Zhang, C. 2010. Increasing Supplier-Driven Innovation: When customers collaborate with suppliers they can build trust, reduce relational stress, and increase innovation-related activities. *MIT Sloan Management Review*. Winter (2010), 41–46.
- [22] Jansen, J.J.P. et al. 2012. Ambidexterity and performance in multiunit contexts: Cross-level moderating effects of structural and resource attributes. *Strategic Management Journal*. 33, 11 (Nov. 2012), 1286–1303. DOI:https://doi.org/10.1002/smj.1977.
- [23] Johnsen, T. et al. 2006. Centrality of customer and supplier interaction in innovation. *Journal of Business Research*. 59, 6 (2006), 671–678. DOI:https://doi.org/10.1016/j.jbusres.2005.11.003.
- [24] Kearney, A. et al. 2018. Executive capability for innovation: the Irish seaports sector. *European Journal of Training and Development*. 42, 5–6 (2018), 342–361. DOI:https://doi.org/10.1108/EJTD-10-2017-0081.
- [25] Keskin, H. 2006. Market orientation, learning orientation, and innovation capabilities in SMEs: An extended model. *European Journal of Innovation Management*. 9, 4 (2006), 396–417. DOI:https://doi.org/10.1108/14601060610707849.
- [26] Kohli, A.K. and Jaworski, B.J. 1990. Market Orientation: The Construct, Research Propositions, and Managerial Implications. *Journal of Marketing*. 54, 2 (1990), 1–18. DOI:https://doi.org/10.1177/00224299005400201.
- [27] Lai, H.C. and Shyu, J.Z. 2005. A comparison of innovation capacity at science parks across the Taiwan Strait: The case of Zhangjiang High-Tech Park and Hsinchu Science-based Industrial Park. *Technovation*. 25, 7 (2005), 805–813. DOI:https://doi.org/10.1016/j.technovation.2003.11.004.
- [28] Levinthal, D.A. and March, J.G. 1993. The myopia of learning. *Strategic Management Journal*. 14, 2 S (1993), 95–112. DOI:https://doi.org/10.1002/smj.4250141009.
- [29] Li, G. 2020. The impact of supply chain relationship quality on knowledge sharing and innovation performance: evidence from Chinese manufacturing industry. *Journal of Business and Industrial Marketing*. 36, 5 (2020), 834–848. DOI:https://doi.org/10.1108/JBIM-02-2020-0109.
- [30] Lim, J.-S. et al. 2017. Market orientation, innovation commercialization capability and firm performance relationships: the moderating role of supply chain influence. *Journal of Business & Industrial Marketing*. 32, 7 (Aug. 2017), 913–924. DOI:https://doi.org/10.1108/JBIM-10-2016-0238.
- [31] Liu, G. et al. 2017. Proactive entrepreneurial behaviour, market orientation, and innovation outcomes. *European Journal of Marketing*. 51, 11/12 (Nov. 2017), 1980–2001. DOI:https://doi.org/10.1108/EJM-11-2016-0663.
- [32] Lui, S.S. and Ngo, H.Y. 2012. Drivers and outcomes of long-term orientation in cooperative relationships. *British Journal of Management*. 23, 1 (2012), 80–95. DOI:https://doi.org/10.1111/j.1467-8551.2010.00719.x.
- [33] Maldonado-Guzmán, G. et al. 2019. Innovation capabilities and performance: are they truly linked in SMEs? *International Journal of Innovation Science*. 11, 1 (2019), 48–62. DOI:https://doi.org/10.1108/IJIS-12-2017-0139.
- [34] March, J.G.J.G. 1991. Exploration and Exploitation in Organizational Learning. *Organization Science*. 2, 1 (1991), 71–87. DOI:https://doi.org/10.1287/orsc.2.1.71.
- [35] Morgan, R.E. and Berthon, P. 2008. Market orientation, generative learning, innovation strategy and business performance inter-relationships in bioscience firms. *Journal of Management Studies*. 45, 8 (2008), 1329–1353. DOI:https://doi.org/10.1111/j.1467-6486.2008.00778.x.
- [36] Morgan, R.M. and Hunt, S.D. 1994. The Commitment-Trust Theory of Relationship Marketing. *Journal of Marketing*. 58, July (1994), 20–38.
- [37] Narver, J.C. et al. 2004. Responsive and Proactive Market Orientation and New-Product Success. *Journal of Product Innovation Management*. 21, 5 (2004), 334–347. DOI:https://doi.org/10.1111/j.0737-6782.2004.00086.x.
- [38] Negassi, S. 2004. R&D co-operation and innovation a microeconomic study on French firms. *Research Policy*. 33, 3 (2004), 365–384. DOI:https://doi.org/10.1016/j.respol.2003.09.010.
- [39] Ngo, L.V. and O'Cass, A. 2012. In search of innovation and customer-related performance superiority: The role of market orientation, marketing capability, and innovation capability interactions. *Journal of Product Innovation Management*. 29, 5 (2012), 861–877. DOI:https://doi.org/10.1111/j.1540-5885.2012.00939.x.
- [40] O'Cass, A. and Sok, P. 2013. Exploring innovation driven value creation in B2B service firms: The roles of the manager, employees, and customers in value creation. *Journal of Business Research*. 66, 8 (Aug. 2013), 1074–1084. DOI:https://doi.org/10.1016/j.jbusres.2012.03.004.
- [41] Panayides, P.M. and So, M. 2005. Logistics service provider-client relationships. *Transportation Research Part E: Logistics and Transportation Review*. 41, 3 (2005), 179–200. DOI:https://doi.org/10.1016/j.tre.2004.05.001.
- [42] Pascual-Fernández, P. et al. 2021. Key drivers of innovation capability in hotels: implications on performance. *International Journal of Hospitality Management*. 94, May 2020 (2021). DOI:https://doi.org/10.1016/j.ijhm.2020.102825.
- [43] Sahoo, S. 2019. Quality management, innovation capability and firm performance: Empirical insights from Indian manufacturing SMEs. *TQM Journal*. 31, 6 (2019), 1003–1027. DOI:https://doi.org/10.1108/TQM-04-2019-0092.
- [44] Salavou, H. 2002. Profitability in market-oriented SMEs: Does product innovation matter? *European Journal of Innovation Management*. 5, 3 (2002), 164–171. DOI:https://doi.org/10.1108/14601060210436736.
- [45] Santos-Vijande, M.L. et al. 2013. An assessment of innovativeness in KIBS: Implications on KIBS' co-creation culture, innovation capability, and performance. *Journal of Business and Industrial Marketing*. 28, 2 (2013), 86–102. DOI:https://doi.org/10.1108/08858621311295236.
- [46] Saunila, M. 2016. Performance measurement approach for innovation capability in SMEs. *International Journal of Productivity and Performance Management*. 65, 2 (Feb. 2016), 162–176. DOI:https://doi.org/10.1108/IJPPM-08-2014-0123.
- [47] Sulisty, H. and Siyamtinah. 2016. Innovation capability of SMEs through entrepreneurship, marketing capability, relational capital and empowerment. *Asia Pacific Management Review*. 21, 4 (2016), 196–203. DOI:https://doi.org/10.1016/j.apmr.2016.02.002.
- [48] Szalavetz, A. 2019. Industry 4.0 and capability development in manufacturing subsidiaries. *Technological Forecasting and Social Change*. 145, July 2017 (2019), 384–395. DOI:https://doi.org/10.1016/j.techfore.2018.06.027.
- [49] Teece, D.J. et al. 1997. Dynamic capabilities and strategic management. *Strategic Management Journal*. 18, 7 (1997), 509–533. DOI:https://doi.org/10.1093/0199248540.003.0013.
- [50] Tomlinson, P.R. and Fai, F.M. 2016. The impact of deep vertical supply chain relationships upon focal-firm innovation performance. *R and D Management*. 46, (2016), 277–290. DOI:https://doi.org/10.1111/radm.12181.
- [51] Uotila, J. et al. 2009. Exploration, exploitation, and financial performance: analysis of S&P 500 corporations. *Strategic Management Journal*. 30, 2 (Feb. 2009), 221–231. DOI:https://doi.org/10.1002/smj.738.
- [52] Villena, V.H. et al. 2011. The dark side of buyer-supplier relationships: A social capital perspective. *Journal of Operations Management*. 29, 6 (2011), 561–576. DOI:https://doi.org/10.1016/j.jom.2010.09.001.
- [53] Wang, Y. and Liu, F. 2020. Proactive or responsive market orientation for stronger service innovation capability: the moderating roles of contractual and relational governance. *Journal of Business and Industrial Marketing*. 35, 5 (2020), 863–874. DOI:https://doi.org/10.1108/JBIM-05-2019-0185.
- [54] Yang, C.C. 2012. Assessing the moderating effect of innovation capability on the relationship between logistics service capability and firm performance for ocean freight forwarders. *International Journal of Logistics Research and Applications*. 15, 1 (2012), 53–69. DOI:https://doi.org/10.1080/13675567.2012.669469.
- [55] Zawislak, P.A. et al. 2018. Technological intensity and innovation capability in industrial firms. *Innovation & Management Review*. 15, 2 (2018), 189–207. DOI:https://doi.org/10.1108/inmr-04-2018-012.
- [56] Zhang, M. and Hartley, J.L. 2018. Guanxi, IT systems, and innovation capability: The moderating role of proactiveness. *Journal of Business Research*. 90, April (Sep. 2018), 75–86. DOI:https://doi.org/10.1016/j.jbusres.2018.04.036.
- [57] Zhou, K.Z. et al. 2008. Market Orientation, Job Satisfaction, Product Quality, and Firm Performance: Evidence From China. *Business*. 29, (2008), 985–1000. DOI:https://doi.org/10.1002/smj.